Seyed Ali Padash Sotoodeh

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EDUCATION

BSc in Computer Engineering Sep 2021 – Feb 2026
K. N. Toosi University of Technology, Tehran, Iran
Final grade: **18.37 / 20**
Thesis: *Sentiment Analysis of User Comments Using Artificial Intelligence* (grade: 19.5 / 20)

Secondary School Diploma – Mathematics and Physics Sep 2018 – Jun 2021
Professor Samii High School, Rasht, Iran
Final grade: **19.38 / 20**

WORK EXPERIENCE

Head Teaching Assistant Jul 2025 – Aug 2026
Sejong University, Seoul, South Korea (remote) · Supervisor: Prof. Sadeghi-Niaraki

- Led a team of 6 TAs producing lecture slides, course materials and instructional videos for *Basic Machine Learning* and *Mathematics for Deep Learning*; designed the workflow and reviewed the team's output.
- Automated lecture-video production with a Python-based tool that replaced manual screen recording and typing, cutting production time by roughly one-third.

Teaching Assistant – Fundamentals of Programming Winter semesters 2022/23 and 2024/25
K. N. Toosi University of Technology

- Edited the course's lesson documents on C programming and designed coding exercises.
- Answered course questions and mentored a group of assigned students.

Python Developer (credited university internship) Jul 2024 – Aug 2024
K. N. Toosi University of Technology

- Refactored a Python web crawler into modular, SOLID-based classes with logging, and extended it with multiprocessing, SQL process tracking and a Django + React interface.

PROJECTS

ChamFlick DE: German Learning App for Books and Films (Android) Aug 2026 – present
ali-padash.github.io/chamflick-showcase

- Building an offline Android app for learning German from books and films: tap any word in a PDF, EPUB, scanned page or live subtitle to open a dictionary card with its meaning, grammar and pronunciation, all on-device.
- Colour-codes nouns by gender (der/die/das/plural), using a sentence parser (spaCy's German model, run on the phone by a C++ re-implementation) to resolve ambiguous words and leaving uncertain ones uncoloured; lookups rejoin separable verbs (*fing … an* → *anfangen*) and split compounds (*Apfelbaum* → *Apfel* + *Baum*).

YouTube Comment Sentiment Analysis (BSc Thesis) Jun 2025 – Oct 2025

github.com/ali-padash/youtube-comment-sentiment-analysis

- Studied how closely lightweight classifiers (Logistic Regression, Linear SVM) can imitate a large transformer (Twitter-RoBERTa) on 3-class YouTube comment sentiment: raised their macro F1 against the transformer's labels from 66.4% / 65.1% to about 72.2%.
- Combined VADER lexicon features with weakly supervised data: YouTube comments collected via the YouTube Data API and pseudo-labelled with Twitter-RoBERTa, which outperformed paraphrase augmentation with Gemma 3.
- Built a Streamlit app that labels comments from the API or a CSV, with a hybrid mode that runs the fast model and falls back to Twitter-RoBERTa only when its confidence is low.

Rahi: Bus Ticket Reservation System (Full-Stack Web App) Mar 2024 – May 2024; rebuilt Oct 2026

github.com/ali-padash/bus-ticket-reservation-system

- Built a bus booking site modelled on an Iranian booking service, with trip search and a seat map: React + TypeScript frontend, Go backend (Gin, GORM) on PostgreSQL.
- Rebuilt in Oct 2026 with a layered REST API, ticket booking with a 10-minute seat hold, a bilingual English/Persian mobile-friendly interface, CI and Docker.
- Prevented double-booking with row-level locks and a partial unique index, verified by concurrent tests (20 simultaneous buyers for one seat, exactly one succeeds).

Further projects

8-bit CPU · Verilog, ModelSim

Reliable transfer over UDP · Go-Back-N, CRC

Genetic algorithm in hardware · SystemC

Crack segmentation · U-Net in PyTorch, own dataset

Details and more at ali-padash.github.io

LANGUAGES

Persian: Native

English: TOEFL iBT 5.5 / 6 (110 / 120), CEFR C1 (Jun 2026)

German: B1

DIGITAL SKILLS

Programming:	Python, Java, Kotlin, Go, TypeScript/JavaScript, C, SQL
ML & NLP:	scikit-learn, PyTorch, Transformers, spaCy, pandas, NumPy, OpenCV
Web & Mobile:	Django, React, Node.js/Express, Gin, PostgreSQL, Streamlit, Android
Hardware Design:	Verilog, VHDL, SystemC, ModelSim
Tools:	Git, GitHub Actions, Docker, LaTeX

INTERESTS

Music (composing in FL Studio) · Learning German through films and comics